Catherine Wu

catw2188@gmail.com | (858)-205-7682

EDUCATION

Harvard University **2014 - 2018**
B.A. GPA: 3.89/4.00 Applied Math-Neuroscience, minor in Mind, Brain, Behavior (MBB)

Columbia University **2022 - 2023**
M.A. in Economics, GPA: 3.95

New York University – Stern School of Business **2024 - Present**
Ph.D. Student, Information Systems

RESEARCH

Research with Professor Arun Sundarajan @ NYU **Sep 2024 – Present**
PhD Student

- *Incentives for Digital Twins: Task-Based Productivity Enhancements with Generative AI*: https://arxiv.org/abs/2509.08732
- Developed a principal-agent model analyzing the incentives and risks of training personalized AI "twins," exploring how task editability and verifiability shape the trade-off between productivity gains and potential worker displacement
- Wrote a book chapter"The Effect of Generative AI on Work Productivity: A Synthesis of the Early Literature" for Microsoft's *Learning Curves*
- Presented at AI @ Wharton, EC'25 Human-AI Collaboration Workshop, AI and Innovation @ NYU Stern

Forecasting Research Institute **Jun 2024 – Sep 2024**
Research Assistant

- Contributed to a bio- and nuclear-risk forecasting project by developing questions for expert/superforecaster surveys, designing forecast evaluation rubrics, and analyzing results using R
- Conducted research on niche existential risk topics, assessed question quality (e.g., divisiveness), and wrote scripts to analyze forecast data and guide survey design

Research with Professor Yeon-Koo Che @ Columbia University **Feb 2023 – Dec 2023**
Research Assistant

- Investigated the effectiveness of RGGI on emission reductions and long-term technology investments, designing a novel way to measure investments using the marginal abatement cost curve
- Reviewed literature on the design and efficiency of carbon markets, presented key findings weekly

Karr Lab @ Mount Sinai **Jan 2018**
Research Intern

- Developed computational models for chemical reactions in a cell, given inputs and chemical equations, based on ordinary differential equations, as one component of a whole-cell computational model
- Participated in a reading group with other researchers on topics such as computational neuroscience

Salk Institute for Biological Studies **May 2012 – Aug 2013**
Research Intern

- Discovered a previously unexplored link between SUMOylation, RNA polymerase, & associated transcription factors, and investigated the mechanism behind SUMOylation as a regulatory process for rRNA transcription
- *Defective RNA polymerase III is negatively regulated by the SUMO-Ubiquitin-Cdc48 pathway.* Z Wang, C Wu, A Aslanian, JR Yates III, T Hunter. Elife, 2018.

TEACHING

NYU Stern School of Business **Sep 2025 – Present**
Teaching Assistant for Microeconomics I (2025)
Teaching Assistant for IT in Business and Society (2026)

Columbia University **Jun 2023 – Dec 2023**
Teaching Assistant for International Open Macroeconomy @ SIPA
Teaching Assistant for Microeconomics I

Harvard School of Engineering and Applied Sciences **Jan 2015 – May 2017**
Teaching Fellow for CS 134: Networks (2017)
Course Assistant for Math 21: Multivariable Calculus and Linear Algebra (2016)
Course Assistant for Math 145: Math in the World (2015)

Summer Program for Applied Rationality and Cognition (SPARC) **Aug 2014 – June 2024**
Instructor
- Wrote and taught custom classes on topics such as reverse game theory, information theory, negotiations and measuring what matters.

Admissions Lead
- Designed admission process to select top 30 of 300 applicants based on analytical and technical abilities
- Constructed a rubric and format that has been used by admissions committee for the past ten years

USA Biology Olympiad (USABO) **June 2016, June 2017, June 2018**
Teaching Assistant
- Trained the top 20 high school biology students in the US in critical thinking in biology research and analysis
- Wrote practical exams in topics including anatomy, ecology and molecular biology, in order to test data analysis skills and experimental laboratory techniques to select the International Biology Olympiad US Teams

WORK EXPERIENCE

Shopify, Data Platform **Apr 2020 - May 2022**
Product Manager
- Championed a pilot to analyze adoption of two open-source BI tools, recruited data scientists to be beta users, monitored usage and usability, worked with engineers to investigate maintenance requirements, developed acceptance criteria
- Built the strategy to increase data-informed decision-making across the company, prioritized the roadmap based on strategy, user feedback and quick wins, created north star metrics for two internal discovery products and built KPI dashboards
- Increased active users on the home page by 30% with further iteration planned based on existing user habits

Mastercard, Data & Services **Aug 2018 - Apr 2020**
Product Manager - Technical II
- Built new diagnostic tool that sped up model iteration process significantly, coordinated with other engineering teams to adapt existing analytical techniques while maintaining consistency of results
- Investigated analytical techniques to diagnose forecast models and identify dataset shift in historical records

Susquehanna International Group (SIG) **May 2017 – Aug 2017**
Quantitative Trading Intern
- Developed a momentum strategy to predict treasury bond pricing, using Python and Excel VBA

- Built a program to scrape upcoming treasury auction data for G8 economies into a calendar and standardize pricing reports

Harvard Association for US-China Relations (HAUSCR) **Aug 2014 – Aug 2017**
President (2017)
HSYLC Director (2016)

- Led a 501c3 non-profit that organized 3 conferences promoting liberal arts education and social impact in Chinese high school students; shaped associate and executive board leadership and professional development
- Directed a 400-student program with a $700,000 budget in China to foster leadership and entrepreneurship, led a team of Chinese and American staff and volunteers, managed internal conflicts and handled crises

AWARDS and ACHIEVEMENTS

Achievements: USAMO qualifier (2013), International Biology Olympiad gold medal (2013), International Linguistics Olympiad silver medal (2014)

SKILLS and INTERESTS

Programming – React/Typescript, Python, SQL, R, Matlab
Languages – heritage speaker in Mandarin, 5 years of French, 2 years of Japanese
Interests – puzzle hunts, crosswords, reading, bouldering